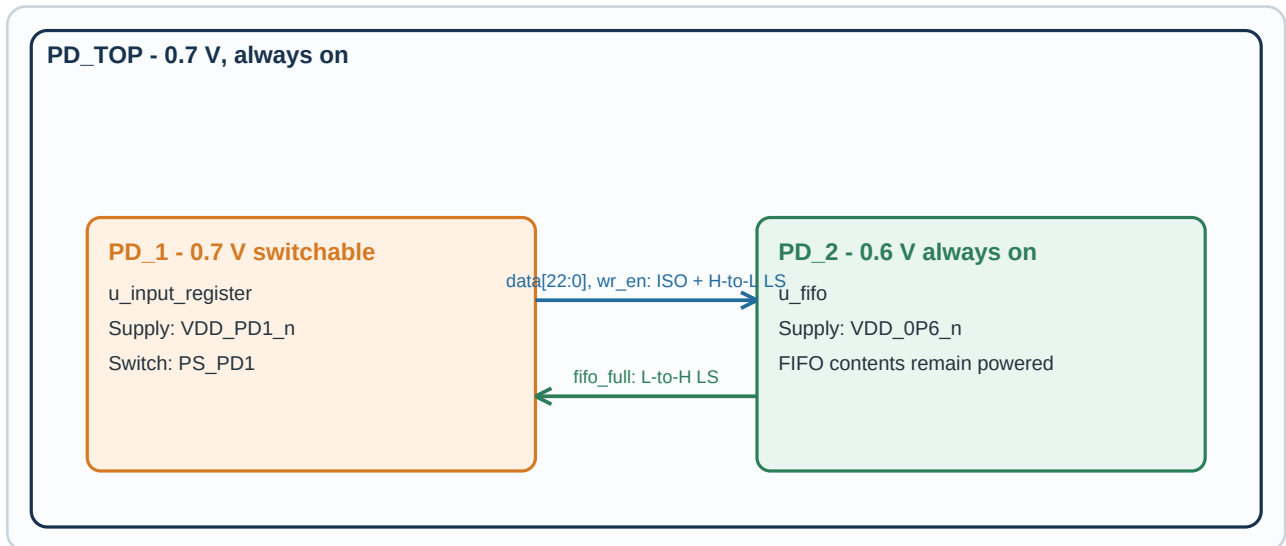


Registered Synchronous FIFO

UPF architecture, implementation notes, corrections, and verification results



Current verified state

The FIFO and input register are assigned to separate voltage domains. PD_1 is a switchable 0.7 V register domain; PD_2 is an always-on 0.6 V FIFO domain. Questa recognizes 25 valid level shifters and 25 valid isolation crossings, with no missing level shifter or isolation warnings in the latest static report.

Scope of these notes

- Explain the intent behind every important UPF command.
- Record the corrections made during Questa integration and why they were needed.
- Provide a clean reference UPF and matching Questa commands.
- Interpret the final static report and define the next dynamic power-off test.

Design: `registered_sync_fifo` | Simulator: QuestaSim 2026.2_2 | UPF semantics: IEEE 1801 UPF 2.1

1. Design context and objective

The registered FIFO wrapper combines two functional blocks. The input register accepts incoming data and presents a registered write transaction to the FIFO. The synchronous FIFO stores the data and supports independent write and read enables under a common clock.

Hierarchy	Function	Power intent
registered_sync_fifo	Top-level wrapper and power-control boundary	PD_TOP, 0.7 V, always on
u_input_register	Registers input data and produces FIFO write data/write enable	PD_1, 0.7 V, switchable
u_fifo	Eight-entry synchronous FIFO with read and write clock gating	PD_2, 0.6 V, always on

Why two domains?

The register is intended to switch off when it is idle, while the FIFO must remain powered so that already stored data is preserved and read-side operation can continue. Different operating voltages create crossings that require level shifting. Turning PD_1 off creates a second requirement: signals leaving PD_1 must be isolated before unknown values can reach PD_2 or other always-on logic.

Functional signals at the internal boundary

Direction	Signal	Width	Required protection
PD_1 -> PD_2	fifo_data_o -> folded_history_i	23	0.7-to-0.6 V level shifting and isolation
PD_1 -> PD_2	fifo_wr_en_o -> wr_en_i	1	0.7-to-0.6 V level shifting and isolation
PD_2 -> PD_1	fifo_full_o -> fifo_full_i	1	0.6-to-0.7 V level shifting
PD_1 -> PD_TOP	data_ready_o boundary	1	Isolation while PD_1 is off

Important model boundary

Power-aware RTL simulation models logic corruption, isolation, power states, and inserted power-management behavior. It does not simulate analog signal amplitude loss as a logic value travels from 0.7 V to 0.6 V. Missing or incorrect level shifting is found by structural power-aware checks.

2. Power domains

```
upf_version 2.1

create_power_domain PD_TOP -include_scope
create_power_domain PD_1 -elements {u_input_register}
create_power_domain PD_2 -elements {u_fifo}
```

PD_TOP uses `-include_scope`, so the current UPF scope - the wrapper selected by `-pa_top` - belongs to PD_TOP. Child instances explicitly placed in PD_1 and PD_2 become lower boundaries of PD_TOP.

PD_1 contains only `u_input_register`. Its internal gated-clock instance inherits PD_1 because it is below that register instance.

PD_2 contains only `u_fifo`. The FIFO read and write clock-gating instances inherit PD_2.

Why the instance names are relative

The `vopt` command applies the UPF at `/registered_sync_fifo_tb/dut`. Therefore, `u_input_register` and `u_fifo` are resolved relative to that DUT scope; the full testbench hierarchy is not written inside the UPF.

Verified architecture

- PD_TOP extent: the `registered_sync_fifo` wrapper, with PD_1 and PD_2 marked as lower boundaries.
- PD_1 extent: `u_input_register`; reported by Questa as switchable.
- PD_2 extent: `u_fifo`; reported by Questa as always on.

Naming convention used in this project

Supply ports retain names such as `VDD_0P7`. Supply nets use an `_n` suffix, such as `VDD_0P7_n`. This was chosen to make ports and nets visually distinct. In many RTL conventions, `_n` means active-low, so this project-specific convention should be documented wherever the UPF is reused.

3. Supply ports, nets, and domain supplies

```
create_supply_port VDD_0P7 -direction in
create_supply_port VDD_0P6 -direction in
create_supply_port VSS -direction in

create_supply_net VDD_0P7_n
create_supply_net VDD_0P6_n
create_supply_net VSS_n

connect_supply_net VDD_0P7_n -ports {VDD_0P7}
connect_supply_net VDD_0P6_n -ports {VDD_0P6}
connect_supply_net VSS_n -ports {VSS}

set_domain_supply_net PD_TOP -primary_power_net VDD_0P7_n -primary_ground_net VSS_n
set_domain_supply_net PD_2 -primary_power_net VDD_0P6_n -primary_ground_net VSS_n
```

A **supply port** is an external connection point through which the power-aware model receives a supply. A **supply net** distributes that supply inside the modeled design. `connect_supply_net` connects the external port to its internal distribution net.

`set_domain_supply_net` assigns the primary power and ground functions used by a domain. PD_TOP receives 0.7 V directly, and PD_2 receives 0.6 V directly. Both use the same ground net.

Supply mapping

Domain	Primary power	Ground	Behavior
PD_TOP	VDD_0P7_n	VSS_n	Always on at 0.7 V
PD_1	VDD_PD1_n	VSS_n	Switched 0.7 V supply
PD_2	VDD_0P6_n	VSS_n	Always on at 0.6 V

Testbench supply control

```
supply_on("VSS", 0.0);
supply_on("VDD_0P7", 0.7);
supply_on("VDD_0P6", 0.6);
```

These calls drive the UPF-created supply ports during simulation. They establish the external rails; they do not directly bypass the PD_1 power switch.

4. Switchable supply for PD_1

```
create_supply_net VDD_PD1_n -domain PD_1

create_power_switch PS_PD1 \
-domain PD_1 \
-input_supply_port {VIN VDD_0P7_n} \
-output_supply_port {VOUT VDD_PD1_n} \
-control_port {CTRL pd1_power_on_i} \
-on_state {PD1_SW_ON VIN {CTRL}} \
-off_state {PD1_SW_OFF {!CTRL}}

set_domain_supply_net PD_1 -primary_power_net VDD_PD1_n -primary_ground_net VSS_n
```

The switch input port VIN is associated with the always-on 0.7 V net. The switch output port VOUT is associated with the switched net VDD_PD1_n. The output-supply-port definition performs the supply-net association; a separate connect_supply_net is not required for this switch output.

Switch behavior

pd1_power_on_i	Switch state	VDD_PD1_n	Expected PD_1 behavior
1	PD1_SW_ON	FULL_ON at 0.7 V	Register operates normally
0	PD1_SW_OFF	OFF	Nonretained register state is corrupted

Control sequencing requirement

Isolation must be asserted before pd1_power_on_i is driven low. During startup, power must be restored and the nonretained register reset before isolation is released.

Why the power switch is needed

Assigning PD_1 directly to VDD_0P7_n would make it always on. The modeled switch creates a distinct switched rail and lets Questa determine when PD_1 is powered, off, or corrupt. It also gives static and dynamic checks a defined control signal and switch state.

5. Supply-port states and domain power states

```
add_port_state VDD_0P7 -state {VDD_0P7_ON 0.7}
add_port_state VDD_0P6 -state {VDD_0P6_ON 0.6}
add_port_state VSS -state {VSS_ON 0.0}
```

Port states associate names with electrical values on the external supply ports. This provides the voltage information used when Questa determines whether a crossing is high-to-low or low-to-high.

```
add_power_state PD_TOP.primary \
-state {TOP_ON \
-supply_expr {power == {FULL_ON 0.7} && ground == {FULL_ON 0.0}} \
-simstate NORMAL}

add_power_state PD_1.primary \
-state {PD1_ON \
-supply_expr {power == {FULL_ON 0.7} && ground == {FULL_ON 0.0}} \
-simstate NORMAL} \
-state {PD1_OFF \
-supply_expr {power == OFF && ground == {FULL_ON 0.0}} \
-simstate CORRUPT}

add_power_state PD_2.primary \
-state {PD2_ON \
-supply_expr {power == {FULL_ON 0.6} && ground == {FULL_ON 0.0}} \
-simstate NORMAL}
```

`add_power_state` names and interprets electrical conditions. It does not itself turn a supply on or off. `PD_1` has both a normal operating state and an off state. The `CORRUPT` simulation state tells the power-aware simulator that ordinary state in the unpowered, nonretained domain cannot be trusted.

Why power states were added here

- The design has more than one voltage and more than one operating condition.
- Static crossing checks need clear voltage relationships for level-shifter direction.
- Dynamic simulation needs a defined corrupt state when `PD_1` is switched off.
- Named states make reports and future legal-mode checks easier to understand.

Difference from earlier simple UPF runs

A single-domain, always-on design can often simulate using only supply definitions and `supply_on` calls. This design benefits from explicit power states because it contains a switched domain, voltage crossings, isolation, and multiple legal operating conditions.

6. Level-shifter strategies

Level shifters are selected by the source and sink supply sets. The rule describes the required voltage direction. The domain and applies_to options identify the boundary where the strategy is evaluated.

```
set_level_shifter LS_PD1_TO_PD2 \
-domain PD_2 \
-applies_to inputs \
-source PD_1.primary \
-sink PD_2.primary \
-rule high_to_low \
-location self

set_level_shifter LS_PD2_TO_PD1 \
-domain PD_1 \
-applies_to inputs \
-source PD_2.primary \
-sink PD_1.primary \
-rule low_to_high \
-location self
```

Strategy interpretation

Strategy	Crossing	Rule	Placement meaning
LS_PD1_TO_PD2	PD_1 0.7 V -> PD_2 0.6 V	high_to_low	self: at the receiving PD_2 input boundary
LS_PD2_TO_PD1	PD_2 0.6 V -> PD_1 0.7 V	low_to_high	self: at the receiving PD_1 input boundary

Why source and sink are explicit

-applies_to inputs alone can select every input entering a domain. Adding -source and -sink restricts the strategy to the intended supply-to-supply relationship. This prevents unrelated same-voltage inputs from being selected.

Meaning of location

- **self**: place the inserted cell within the strategy domain, at that domain's boundary.
- **parent**: place the cell outside the strategy domain, in its parent domain.

Correction made

The first reverse strategy used PD_TOP as the source and PD_2 as the sink while declaring a low-to-high rule. That did not describe fifo_full_o traveling from PD_2 to PD_1, so Questa reported one LS_MISSING crossing. The corrected strategy uses source PD_2.primary and sink PD_1.primary. The rerun reports one valid low-to-high level shifter for fifo_full_o.

7. Isolation strategies

```

set_isolation ISO_PD1_TO_PD2 \
-domain PD_1 \
-applies_to outputs \
-source PD_1.primary \
-sink PD_2.primary \
-isolation_supply_set PD_TOP.primary \
-clamp_value 0 \
-isolation_signal pd1_isolation_i \
-isolation_sense high \
-location parent

set_isolation ISO_PD1_TO_PDTOP \
-domain PD_1 \
-applies_to outputs \
-source PD_1.primary \
-sink PD_TOP.primary \
-isolation_supply_set PD_TOP.primary \
-clamp_value 0 \
-isolation_signal pd1_isolation_i \
-isolation_sense high \
-location parent

```

When `pd1_isolation_i` is high, outputs selected by these strategies are clamped to zero. The isolation cells use `PD_TOP.primary`, an always-on supply, so they remain functional after `PD_1` loses power. Location `parent` places them outside the switchable domain.

Why clamp to zero?

- A zero write enable prevents a spurious FIFO write.
- Zero data provides a known value rather than X at the FIFO boundary.
- The 23 data bits reset to zero, so the report marks their clamp as matching reset.

Why there are two isolation strategies

The original strategy covered `PD_1`-to-`PD_2` crossings. Questa then reported `data_ready_o` as a `PD_1`-to-`PD_TOP` crossing with missing isolation. A second source/sink-specific strategy was added for that always-on destination.

Isolation and level shifting on the same path

For `PD_1`-to-`PD_2` traffic, both functions are necessary: isolation protects the always-on FIFO while `PD_1` is off, and level shifting handles the 0.7 V to 0.6 V voltage difference. At implementation time, the library may provide separate cells or a combined isolation plus level-shifter cell.

8. Consolidated UPF reference

This listing reflects the structure accepted by the successful Questa run and the corrected strategies shown in the final static report.

```
upf_version 2.1

create_power_domain PD_TOP -include_scope
create_power_domain PD_1 -elements {u_input_register}
create_power_domain PD_2 -elements {u_fifo}

create_supply_port VDD_0P7 -direction in
create_supply_port VDD_0P6 -direction in
create_supply_port VSS -direction in

create_supply_net VDD_0P7_n
create_supply_net VDD_0P6_n
create_supply_net VSS_n

connect_supply_net VDD_0P7_n -ports {VDD_0P7}
connect_supply_net VDD_0P6_n -ports {VDD_0P6}
connect_supply_net VSS_n -ports {VSS}

set_domain_supply_net PD_TOP -primary_power_net VDD_0P7_n -primary_ground_net VSS_n
set_domain_supply_net PD_2 -primary_power_net VDD_0P6_n -primary_ground_net VSS_n

create_supply_net VDD_PD1_n -domain PD_1
create_power_switch PS_PD1 \
-domain PD_1 \
-input_supply_port {VIN VDD_0P7_n} \
-output_supply_port {VOUT VDD_PD1_n} \
-control_port {CTRL pd1_power_on_i} \
-on_state {PD1_SW_ON VIN {CTRL}} \
-off_state {PD1_SW_OFF {!CTRL}}
set_domain_supply_net PD_1 -primary_power_net VDD_PD1_n -primary_ground_net VSS_n

add_port_state VDD_0P7 -state {VDD_0P7_ON 0.7}
add_port_state VDD_0P6 -state {VDD_0P6_ON 0.6}
add_port_state VSS -state {VSS_ON 0.0}

add_power_state PD_TOP.primary \
-state {TOP_ON \
-supply_expr {power == {FULL_ON 0.7} && ground == {FULL_ON 0.0}} \
-simstate NORMAL}

add_power_state PD_1.primary \
-state {PD1_ON \
-supply_expr {power == {FULL_ON 0.7} && ground == {FULL_ON 0.0}} \
-simstate NORMAL} \
-state {PD1_OFF \
-supply_expr {power == OFF && ground == {FULL_ON 0.0}} \
-simstate CORRUPT}

add_power_state PD_2.primary \
-state {PD2_ON \
-supply_expr {power == {FULL_ON 0.6} && ground == {FULL_ON 0.0}} \
-simstate NORMAL}
```

8. Consolidated UPF reference - continued

The remaining commands define the level-shifter and isolation strategies for the internal domain crossings.

```
set_level_shifter LS_PD1_TO_PD2 \  
-domain PD_2 -applies_to inputs \  
-source PD_1.primary -sink PD_2.primary \  
-rule high_to_low -location self  
  
set_level_shifter LS_PD2_TO_PD1 \  
-domain PD_1 -applies_to inputs \  
-source PD_2.primary -sink PD_1.primary \  
-rule low_to_high -location self  
  
set_isolation ISO_PD1_TO_PD2 \  
-domain PD_1 -applies_to outputs \  
-source PD_1.primary -sink PD_2.primary \  
-isolation_supply_set PD_TOP.primary \  
-clamp_value 0 -isolation_signal pd1_isolation_i \  
-isolation_sense high -location parent  
  
set_isolation ISO_PD1_TO_PDTOP \  
-domain PD_1 -applies_to outputs \  
-source PD_1.primary -sink PD_TOP.primary \  
-isolation_supply_set PD_TOP.primary \  
-clamp_value 0 -isolation_signal pd1_isolation_i \  
-isolation_sense high -location parent
```

9. Questa compilation and power-aware run

Compile SystemVerilog after RTL or testbench changes:

```
if {[file isdirectory work]} {vlib work}
vmap work work
vlog -sv -L mtiUPF +incdir+./rtl +incdir+./tb \
./rtl/gated_clk.sv \
./rtl/fifo_input_register.sv \
./rtl/sync_fifo.sv \
./rtl/registered_sync_fifo.sv \
./tb/registered_sync_fifo_tb.sv
```

Optimize with UPF and static checks:

```
vopt registered_sync_fifo_tb \
+acc \
-pa_top /registered_sync_fifo_tb/dut \
-pa_upf ./upf/registered_sync_fifo.upf \
-pa_upfversion=2.1 \
-pa_checks=sl+si \
-pa_lib work \
-pa_genrpt=pa+de \
-o registered_sync_fifo_tb_pa_opt
```

Simulate the optimized power-aware design:

```
vsim -sv_seed 12345 registered_sync_fifo_tb_pa_opt -pa -pa_lib work
run 46ns
power add -r /registered_sync_fifo_tb/dut/*
run -all
power report -all -bsaif reports/saif/registered_sync_fifo.saif
pa report -scope=/registered_sync_fifo_tb/dut
```

Meaning of selected options

Option	Purpose
-pa_top	Identifies the DUT hierarchy where the UPF is applied.
-pa_upf	Selects the UPF file; Questa does not choose one automatically.
-pa_upfversion=2.1	Selects UPF 2.1 syntax and semantics. The equals sign is required.
-pa_checks=sl+si	Runs static level-shifter and isolation checks.
-pa_genrpt=pa+de	Generates power architecture and design-element reports.
-o	Names the optimized design consumed by vsim.

10. Problems encountered and corrections

Observed problem	Cause	Correction
Comments after UPF commands were parsed as arguments	A comment followed a command on the same line or after a continuation	Put comments on separate lines; make the backslash the final character of a continued line.
vopt-1928: optimized design name required	The vopt command ended before the -o option because a continuation was missing	Terminate each continued line with backslash and include -o registered_sync_fifo_tb_pa_opt.
vopt-19: failed to access library '2'	-pa_upfversion 2.1 was passed as separate tokens	Use the required form -pa_upfversion=2.1.
add_power_state parsing failed	UPF version semantics were not selected consistently	Declare upf_version 2.1 and invoke vopt with -pa_upfversion=2.1.
One LS_MISSING warning	The PD_2-to-PD_1 strategy had the wrong source and sink	Use source PD_2.primary, sink PD_1.primary, rule low_to_high.
One ISO_MISSING warning	data_ready_o crossed from switchable PD_1 to always-on PD_TOP	Add ISO_PD1_TO_PDTOP using the always-on PD_TOP supply.

Useful Tcl and UPF formatting rules

- Use # for a complete comment line. UPF/Tcl has no C-style block comment syntax.
- A line-continuation backslash must be the last character on the line.
- Do not place a comment after a continuation backslash.
- A UPF-only change requires rerunning vopt. Recompile with vlog only when RTL or testbench source changes.
- Keep the optimized design name identical between vopt -o and the vsim command.

11. Final static verification result

Check	Count	Meaning
LS_VALID	25	24 PD_1-to-PD_2 crossings plus fifo_full_o from PD_2 to PD_1
ISO_VALID	25	24 PD_1-to-PD_2 crossings plus data_ready_o from PD_1 to PD_TOP
ISO_CLAMP_SAME_AS_RESET	23	The 23 FIFO data bits clamp to the same zero value used at reset
LS_MISSING	Absent	No checked voltage crossing lacks a level-shifter strategy
ISO_MISSING	Absent	No checked crossing from switchable PD_1 lacks isolation

Crossing-by-crossing interpretation

- PD_1 -> PD_2 data: 23 valid high-to-low level shifters and 23 valid isolation cells.
- PD_1 -> PD_2 write enable: one valid high-to-low level shifter and one valid isolation cell.
- PD_2 -> PD_1 FIFO full feedback: one valid low-to-high level shifter.
- PD_1 -> PD_TOP data-ready boundary: one valid isolation cell.

What this proves

The static report proves that the UPF strategies cover the crossings identified by Questa and that their directions and isolation requirements are structurally valid. It does not yet prove that the testbench follows the correct power-down and power-up control sequence at runtime.

Implementation limitation

The current simulation uses inferred power-management cells. Before physical implementation or library-based power estimation, the strategies must be mapped to level-shifter and isolation cells available in the target technology libraries. If ASAP7 does not contain suitable characterized multi-voltage cells, custom models or an appropriate low-power library will be required.

12. Next verification step: dynamic PD_1 shutdown

The next UVM sequence should control `pd1_power_on_i` and `pd1_isolation_i` through the interface and driver. The sequence should preserve the FIFO contents while proving that an unpowered input-register domain cannot inject writes or X values into PD_2.

Phase	Control action	Expected observation
1. Normal operation	<code>power_on=1</code> , <code>isolation=0</code> ; write several values	Register and FIFO operate normally
2. Quiesce	Stop new data and disable register activity	No transaction is partially accepted
3. Isolate	Set <code>pd1_isolation_i=1</code>	PD_1 outputs toward PD_2 and PD_TOP clamp to zero
4. Power down	Set <code>pd1_power_on_i=0</code>	PD_1 internal nonretained state becomes corrupt; FIFO stays powered
5. Observe	Read or inspect FIFO while PD_1 is off	Stored FIFO data remains valid; no unintended write occurs
6. Power up	Set <code>pd1_power_on_i=1</code> and wait for stable power	PD_1 is powered but its old state is not assumed valid
7. Reinitialize	Assert and release reset for the input register	<code>data_q</code> and <code>valid_q</code> return to defined reset values
8. Release isolation	Set <code>pd1_isolation_i=0</code> and resume data	Normal register-to-FIFO traffic resumes

Scoreboard expectations

- Do not enqueue data while the isolated write-enable path is clamped low.
- Retain the reference-model FIFO queue while PD_1 is off because PD_2 remains powered.
- Accept FIFO reads during PD_1 shutdown if the wrapper architecture permits them.
- Treat PD_1 internal X values as expected only during the off interval; X values must not reach protected PD_2 inputs.
- After power-up, require reset before accepting new input-register traffic.

Retention decision

Retention for the FIFO memory remains deferred. With PD_2 defined as always on, the FIFO storage remains powered and does not require retention for the current architecture. Retention becomes relevant only if PD_2 or the memory array is later made switchable.

Checkpoint before continuing

The UPF is structurally clean for the presently modeled crossings. The work should resume by adding power-control fields to the UVM interface, transaction, driver, monitor, and scoreboard, followed by the dynamic shutdown sequence above.